

Predicting Digital Experiment Success Using A/B Testing Metrics

Audience Questions and Responses

1. Why was synthetic data selected instead of a real experimentation dataset?

Synthetic data was selected because detailed experimentation datasets are often proprietary and contain sensitive business or customer information. Publicly available A/B testing datasets with rich behavioral metrics are relatively uncommon. Using synthetic data allowed the project to simulate realistic experimentation scenarios while avoiding privacy and confidentiality concerns. It also provided flexibility to control relationships between variables for modeling purposes.

2. How did target leakage affect the initial machine learning model performance?

Target leakage caused the initial models to achieve unrealistically high accuracy scores because several variables directly encoded information tied to the final experiment outcome. Instead of learning generalizable predictive patterns, the models were effectively memorizing the answer. After removing leakage-related variables such as conversion rate outcomes and statistical significance indicators, model performance decreased substantially and produced more realistic evaluation results.

3. Why were Logistic Regression and Random Forest selected for this project?

Logistic Regression was selected because it provides a simple and interpretable baseline classification model. It allowed the project to evaluate how individual metrics related to experiment outcomes. Random Forest was selected because it can capture non-linear relationships and interactions among variables while also supporting feature importance analysis. Comparing both models helped evaluate whether more complex ensemble methods improved predictive performance.

4. Which engagement metrics appeared most influential during feature importance analysis?

After target leakage correction, click-through rate, bounce rate, session duration, and traffic-related engagement metrics demonstrated the strongest remaining influence on

model behavior. These findings suggested that user engagement behavior may provide more predictive value than structural experiment characteristics such as device type alone.

5. Why did the models struggle to identify successful experiments consistently?

The models struggled because experiment success is influenced by many interacting behavioral and contextual factors that were not fully captured within the dataset. Additionally, successful experiments represented a smaller portion of the data, making prediction more difficult. Even after leakage correction, the remaining features provided only moderate predictive signal, which resulted in very low recall scores for successful outcomes.

6. How did class imbalance affect model performance?

Class imbalance caused the models to favor predicting unsuccessful experiments because those outcomes appeared more frequently within the dataset. As a result, overall accuracy remained relatively high, but recall for successful experiments became very low. This demonstrated how accuracy alone can sometimes provide a misleading impression of model effectiveness in imbalanced classification problems.

7. What are the risks of relying too heavily on predictive experimentation systems?

Over-reliance on predictive systems may cause organizations to prioritize only experiments that appear statistically safe or easily predictable, potentially discouraging innovative or long-term strategic experiments. Predictive models may also inherit biases or limitations from the underlying data. Because of this, predictive analytics should support human decision-making rather than replace experimentation expertise and business judgment.

8. How might real-world experimentation data change the results of this analysis?

Real-world experimentation data would likely introduce greater variability, noise, and behavioral complexity compared to the synthetic dataset used in this project. Additional factors such as seasonality, marketing campaigns, user segmentation, and changing business conditions could influence experiment outcomes. As a result, model

performance might decrease initially, but the analysis would likely become more realistic and operationally valuable.

9. What additional variables could improve prediction performance in future versions of the project?

Future versions of the project could incorporate variables such as user segmentation, experiment duration, traffic source, device operating system, time-of-day activity, customer demographics, historical experimentation performance, and seasonal trends. Richer behavioral and contextual variables may improve the models' ability to identify successful experiments more consistently.

10. How could predictive analytics be integrated into real experimentation workflows without replacing human decision-making?

Predictive analytics could function as a decision-support tool that helps experimentation teams prioritize tests, estimate resource allocation, and identify potentially high-impact experiments. However, final experimentation decisions should still involve human oversight, business context, ethical review, and strategic judgment. The goal should be to augment experimentation workflows rather than fully automate them.